

Shifting the Parent Functions

Name: _____

Period: ____

Date: Aug 24

FINAL ANSWERS — RECORD HERE (GRADED ON THE ANSWER; SHOW YOUR WORK)

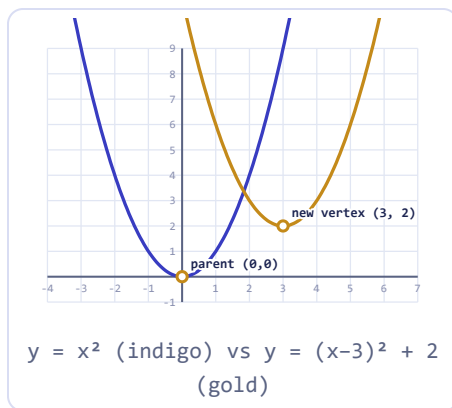
1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____ 10. _____ 11. _____ 12. _____

Use the two shift rules: outside moves as written, inside moves opposite. Domains and ranges slide with the graph.

TYPE 1 Describe the shift

EXAMPLE (DESCRIBE THE SHIFT)

$y = (x - 3)^2 + 2$ — inside $x - 3$ moves RIGHT 3, outside $+2$ moves UP 2. The vertex lands at $(3, 2)$.



Directions: Name the parent, then describe every shift.

1. $y = x^2 + 3$

2. $y = \sqrt{x-4}$

3. $y = |x + 2| - 5$

4. $y = \frac{1}{x} + 6$

TYPE 2 Write the equation

WHY IT WORKS

Reverse the rules: the described move tells you exactly what to write inside (opposite) and outside (as written).

Directions: Write the equation for each described shift.

5. $y = x^3$ shifted down 7

6. $y = |x|$ shifted right 3

7. $y = \frac{1}{x}$ shifted right 4 and down 3

8. $y = \sqrt{x}$ shifted left 5 and up 2

TYPE 3 Domain and range after the shift

WHY IT WORKS

The domain and range slide exactly as far as the graph does.

Directions: State the domain AND range of each shifted function.

9. $y = \sqrt{x-4}$

11. $y = |x+2| - 5$

10. $y = \frac{1}{x-4} - 3$

CHALLENGE Stretch problems (optional)

12. The point $(1, 1)$ sits on $y = \sqrt{x}$. Exactly where does it land on $y = \sqrt{x+5} + 2$? Explain both moves.